

## Assignment: Deterministic Decision Tree

### 1. Introduction

A decision tree is a model used in Artificial Intelligence to make decisions based on conditions. It has a tree-like structure where each node represents a decision and branches represent possible outcomes.

A **deterministic decision tree** always produces the same output for the same input, without any randomness.

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### 2. Objective

To design a deterministic decision tree to predict whether a student will pass or fail based on study hours and attendance.

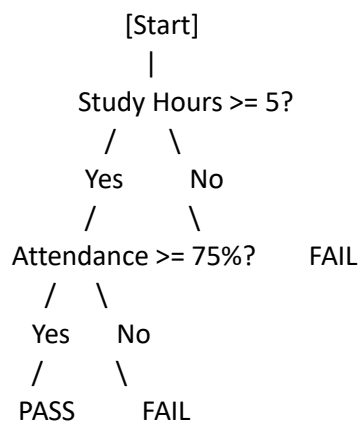
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### 3. Problem Statement

Predict student performance (PASS/FAIL) using:

- Study Hours
  - Attendance Percentage
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### 4. Decision Tree Diagram



### 5. Algorithm

1. Start
2. Check if Study Hours  $\geq 5$ 
  - If No  $\rightarrow$  FAIL
3. If Yes, check Attendance  $\geq 75\%$ 
  - If Yes  $\rightarrow$  PASS

- If No → FAIL

#### 4. Stop

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### 6. Implementation (Python Code)

```
def decision_tree(study_hours, attendance):  
    if study_hours >= 5:  
        if attendance >= 75:  
            return "PASS"  
        else:  
            return "FAIL"  
    else:  
        return "FAIL"  
  
print(decision_tree(6, 80))
```

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### 7. Explanation

The model first checks study hours. If they are less than 5, the student fails. If sufficient, attendance is evaluated. Only students meeting both conditions pass.

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### 8. Advantages

- Simple and easy to understand
  - Fast decision-making
  - Predictable results
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### 9. Disadvantages

- Not suitable for complex problems
  - Requires manual rules
  - Less flexible than advanced AI models
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### 10. Conclusion

Deterministic decision trees provide a simple and reliable method for decision-making in basic AI applications.

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